



Windows Server 2019 Provisioning & Initial Server Administration

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Introduction

Windows Server 2019 is a potent server operating system that is frequently used in commercial settings to handle infrastructure services, network resources, users, and security. It gives administrators the resources they need to install, set up, keep an eye on, and maintain dependable IT systems.

Objective

obtaining hands-on experience setting up and managing a virtual machine running Windows Server 2019. Installing Windows Server 2019 (Desktop Experience), configuring the server initially, enabling remote administration, modifying the server environment, installing necessary Windows features like BitLocker, using Event Viewer to examine system events, and resolving common problems with audio, Remote Desktop, and network connectivity were all part of the implementation.

Theory Topics

Windows Server is Microsoft's enterprise server platform that enables organizations to run and secure applications, services, and workloads across on-premises, hybrid, and cloud environments.

Windows Server Installation Modes:

Desktop Experience contains a GUI, rendering the machine comparatively user-friendly and easier to manage either remotely or locally.

Server Core is a strapped down version of Windows Server. Generally speaking, it contains no GUI.

Windows Server Troubleshooting Basics:

Troubleshooting involves a methodical approach combining native diagnostic tools (like Event Viewer and Performance Monitor), network utilities, and command-line diagnostics.

Server Manager Overview:

Server Manager is a centralized management console in Windows Server that enables IT professionals to provision and manage both local and remote Windows-based servers from their desktops.

BitLocker Drive Encryption:

A built-in Windows security feature that encrypts entire hard drives to protect your data from unauthorized access.

Windows Event Viewer:

A built-in Microsoft tool that records detailed logs of system activities, errors, and application crashes.

Windows Device Manager:

A built-in Windows utility that lets you view, manage, and troubleshoot all the hardware connected to your computer.

Software Requirements

Software: VM workstation

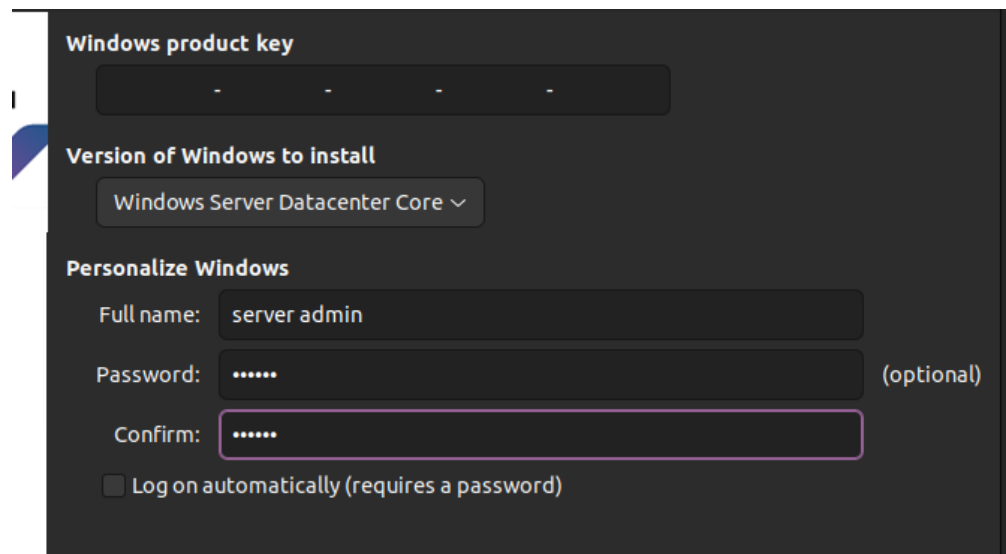
Environment: Two Windows 10 VMs

Access Level: Administrator Account

Implementation

Windows Server Deployment

Standard Core = 2 or less VMs – Data-center Code = multiple VMs, enterprise environment

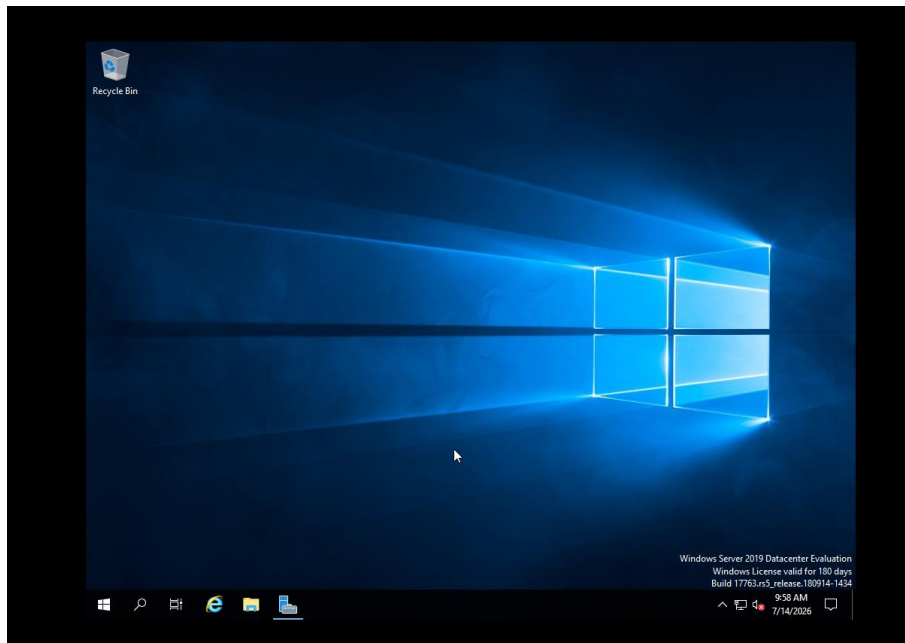


The screenshot shows the Windows Server installation setup interface. It includes fields for the Windows product key, a dropdown menu for the version of Windows to install (set to 'Windows Server Datacenter Core'), and a 'Personalize Windows' section. This section contains input fields for the full name ('server admin'), password (masked with dots), and password confirmation (also masked with dots). There is an '(optional)' label next to the password fields and a checkbox for 'Log on automatically (requires a password)' which is currently unchecked.

Select the operating system you want to install

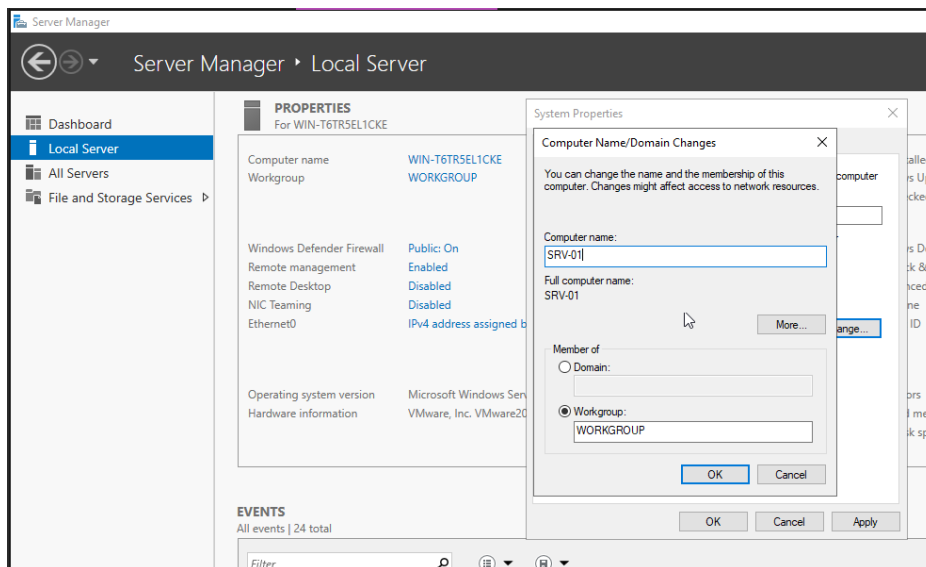
Operating system	Architecture	Date mo
Windows Server 2019 Standard Evaluation	x64	11/5/202
Windows Server 2019 Standard Evaluation (Desktop Experience)	x64	11/5/202
Windows Server 2019 Datacenter Evaluation	x64	11/5/202
Windows Server 2019 Datacenter Evaluation (Desktop Experience)	x64	11/5/202

Description:
This option installs the full Windows graphical environment, consuming extra drive space. It can be useful if you want to use the Windows desktop or have an app that requires it.



Initial Server Configuration

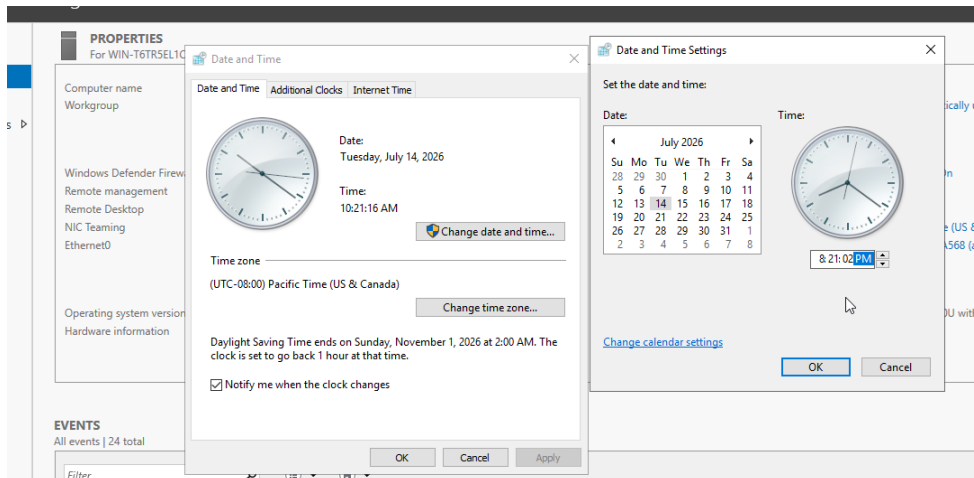
computer name



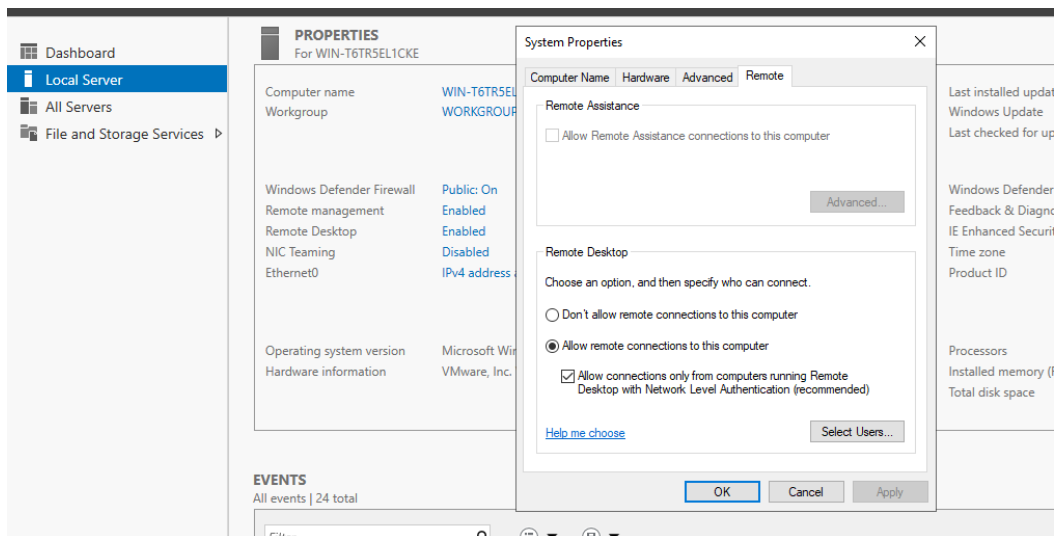
System information

Operating system version	Microsoft Windows Server 2019 Datacenter Evaluation	Processors	AMD Ryzen 5 PRO 6650U with Radeon Graphics
Hardware information	VMware, Inc. VMware20,1	Installed memory (RAM)	4 GB
		Total disk space	59.4 GB

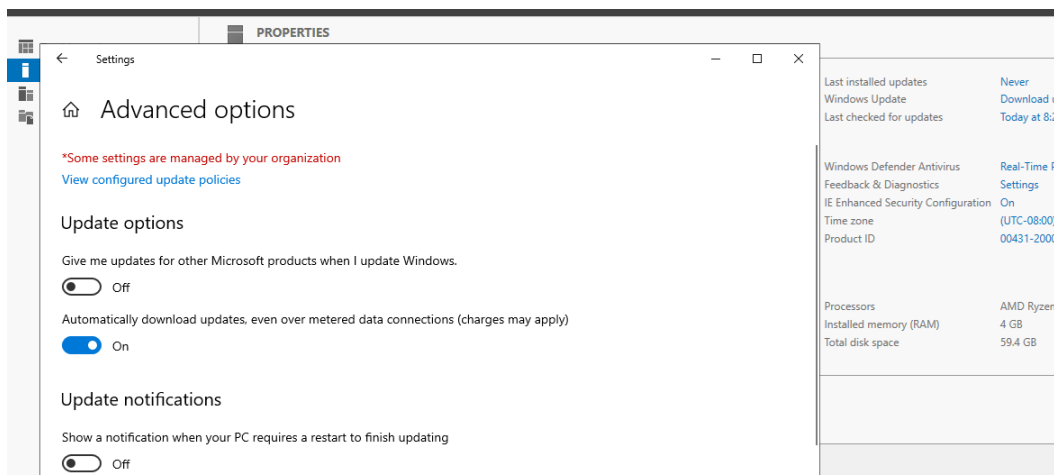
Date & time – Timezone



Remote desktop



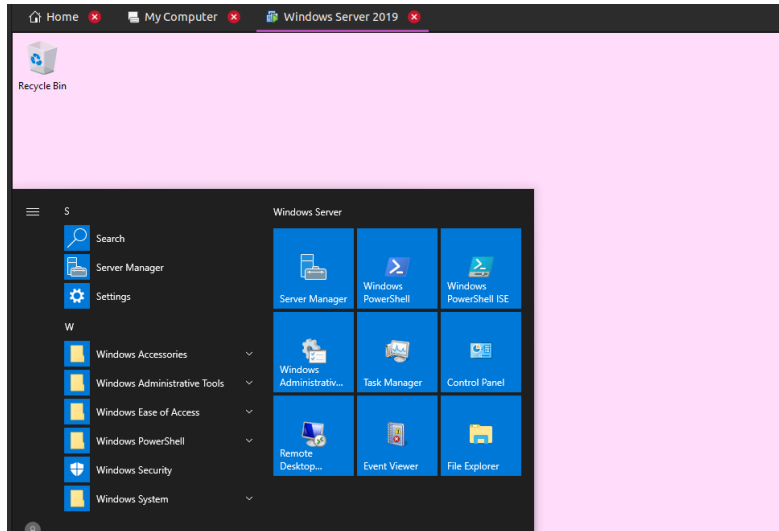
Automatic updates



Windows activation status

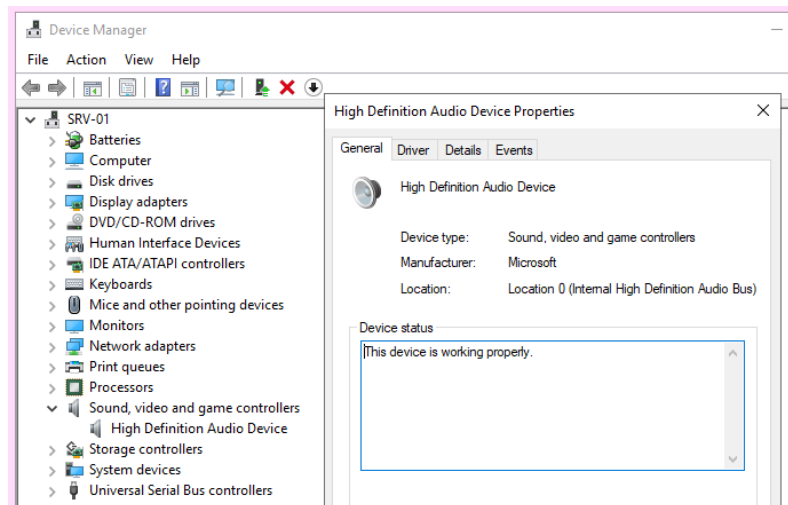
Product ID	00431-20000-00000-AA568 (activated)
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Windows Server Customization

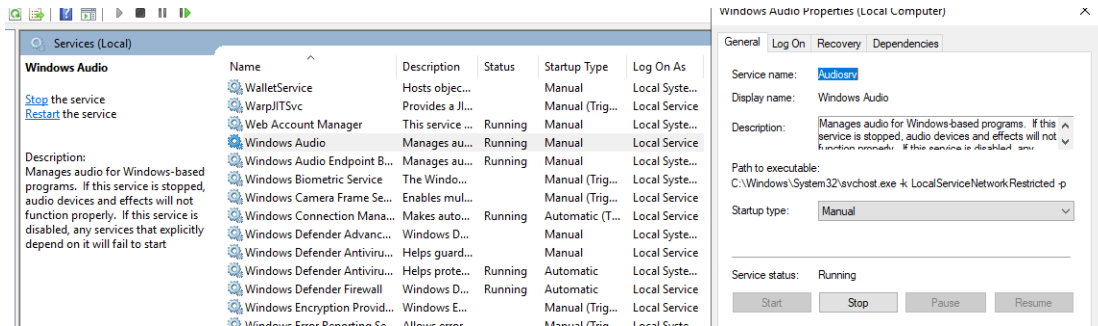


Basic Troubleshooting

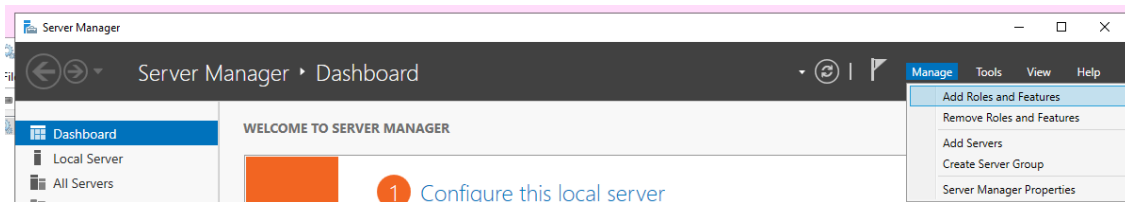
1. Device manager



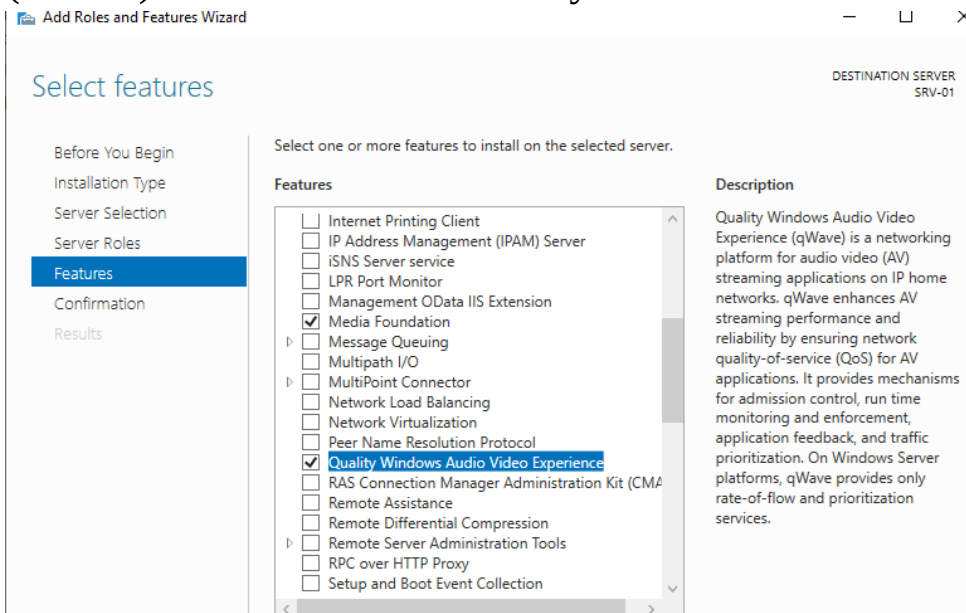
win + R -> services.msc (or shift ctrl esc -> new service -> services.msc)



installing Windows Audio feature (ensures that the system components are correctly registered and will persist through updates or reboots) -> server manager -> manage -> add roles and features

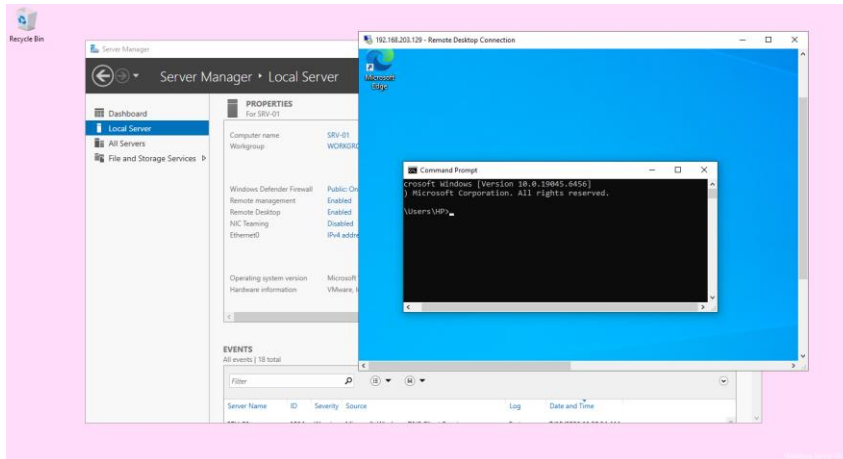


role-based or feature-based installation -> select from server pool (hard disk option is for advanced scenarios where you want to modify a disk file while it is offline) -> file & storage services (default) -> choose and install any media/audio features wanted



*why audio devices are often disabled in server environments?
to reduces unnecessary resource usage, and minimizes the
system's attack surface (because audio devices servers are
designed to provide network services rather than multimedia
functionality)*

2. Remote Desktop Trouble shooting



3. Network connectivity

ipconfig: display the IP address, subnet mask and default gateway for each adapter bound to TCP/IP

```
Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::204c:8130:88bf:b759%5
    IPv4 Address. . . . . : 192.168.203.130
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.203.2
```

ping : determines if a destination device is powered on, connected to the network, and actively responding

```

C:\Users\Administrator>ping 192.168.203.128

Pinging 192.168.203.128 with 32 bytes of data:
Reply from 192.168.203.130: Destination host unreachable.
Reply from 192.168.203.130: Destination host unreachable.
Reply from 192.168.203.130: Destination host unreachable.
Reply from 192.168.203.130: Destination host unreachable.

Ping statistics for 192.168.203.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\Administrator>ping google.com

Pinging google.com [142.251.142.78] with 32 bytes of data:
Reply from 142.251.142.78: bytes=32 time=44ms TTL=128
Reply from 142.251.142.78: bytes=32 time=44ms TTL=128
Reply from 142.251.142.78: bytes=32 time=60ms TTL=128
Reply from 142.251.142.78: bytes=32 time=48ms TTL=128

Ping statistics for 142.251.142.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 60ms, Average = 49ms

```

tracert: displays every router (hop) your data packet passes through to reach the destination. This helps you see where a connection might be dropping or slowing down.

```

Tracing route to google.com [142.251.142.78]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    192.168.203.2
  1  15 ms     2 ms     2 ms     _gateway [192.168.1.1]
  2  23 ms     7 ms     8 ms     10.45.16.60
  3  23 ms     8 ms     6 ms     10.29.27.10
  4  22 ms     7 ms     7 ms     10.37.93.42
  5  22 ms     7 ms     8 ms     10.39.13.93
  6  25 ms     13 ms    12 ms    10.39.15.190
  7  23 ms     11 ms    11 ms    10.45.28.77
  8  27 ms     10 ms    10 ms    81.52.188.108
  9  *         *        *        Request timed out.
 10  558 ms    510 ms   *        72.14.203.56
 11  44 ms     44 ms    48 ms    192.178.105.93
 12  55 ms     44 ms    42 ms    142.251.78.85
 13  60 ms     44 ms    44 ms    ncmrsa-al-in-f14.1e100.net [142.251.142.78]

Trace complete.

C:\Users\Administrator>tracert 192.168.203.128

Tracing route to 192.168.203.128 over a maximum of 30 hops

  0  SRV-01.localdomain [192.168.203.130] reports: Destination host unreachable.

Trace complete.

```

nslookup: queries the Domain Name System (DNS) to identify the specific IP address associated with a domain name.

```

C:\Users\Administrator>nslookup google.com
Server:  UnKnown
Address:  192.168.203.2

Name:     google.com.localdomain
Addresses: 142.251.142.78
           142.251.142.78

C:\Users\Administrator>nslookup facebook.com
Server:  UnKnown
Address:  192.168.203.2

Name:     facebook.com.localdomain
Addresses: 102.132.103.35
           102.132.103.35

```

hostname: local machine name

```

C:\Users\Administrator>hostname
SRV-01

```

testing local connectivity: (testing the internal state of your network card (the "loopback" interface). This will almost always succeed as long as your network adapter is enabled, even if you are completely disconnected from the actual network. It does not verify that you are physically or logically connected to the network infrastructure)

```

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::204c:8130:88bf:b759%5
    IPv4 Address. . . . . : 192.168.203.130
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.203.2

C:\Users\Administrator>ping 192.168.203.130

Pinging 192.168.203.130 with 32 bytes of data:
Reply from 192.168.203.130: bytes=32 time<1ms TTL=128
Reply from 192.168.203.130: bytes=32 time<1ms TTL=128
Reply from 192.168.203.130: bytes=32 time<1ms TTL=128
Reply from 192.168.203.130: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.203.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

testing internet connectivity:

```
C:\Users\Administrator>ping youtube.com

Pinging youtube.com [142.251.36.142] with 32 bytes of data:
Reply from 142.251.36.142: bytes=32 time=72ms TTL=128
Reply from 142.251.36.142: bytes=32 time=46ms TTL=128
Reply from 142.251.36.142: bytes=32 time=46ms TTL=128
Reply from 142.251.36.142: bytes=32 time=49ms TTL=128

Ping statistics for 142.251.36.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 46ms, Maximum = 72ms, Average = 53ms
```

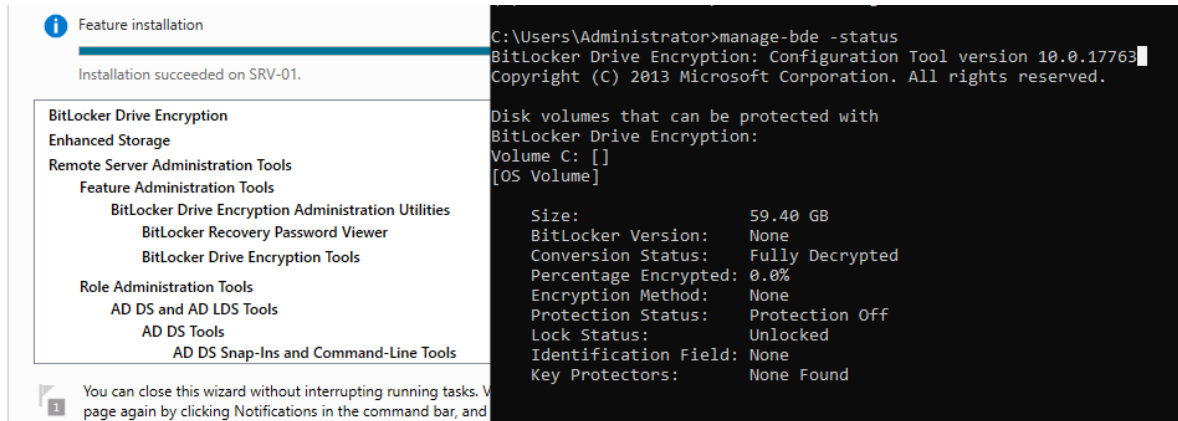
testing communication with another VM:

```
C:\Users\Administrator>ping 192.168.203.129

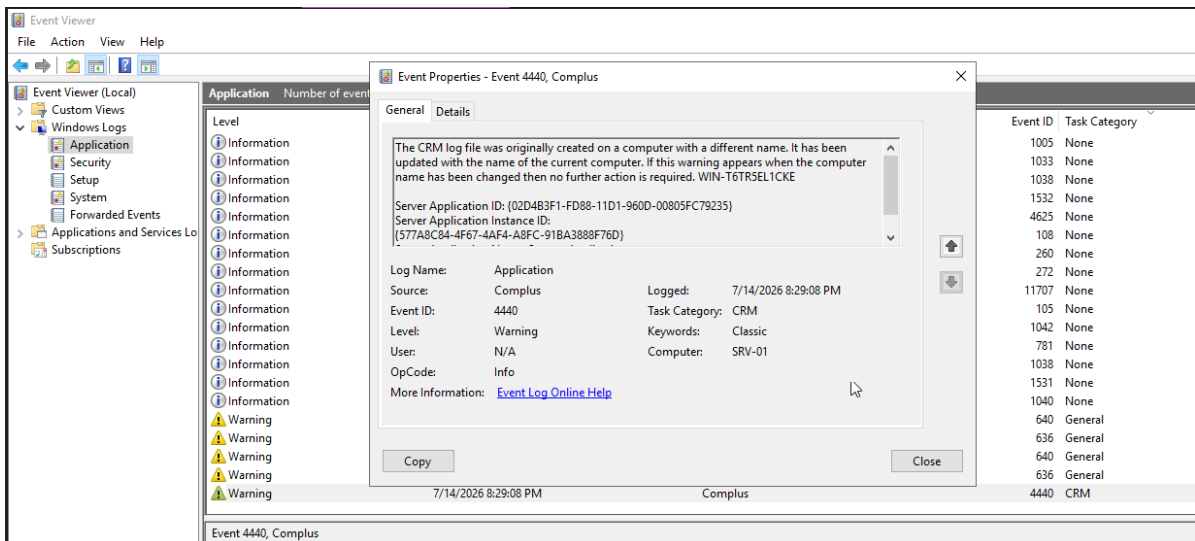
Pinging 192.168.203.129 with 32 bytes of data:
Reply from 192.168.203.129: bytes=32 time<1ms TTL=128
Reply from 192.168.203.129: bytes=32 time<1ms TTL=128
Reply from 192.168.203.129: bytes=32 time<1ms TTL=128
Reply from 192.168.203.129: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.203.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

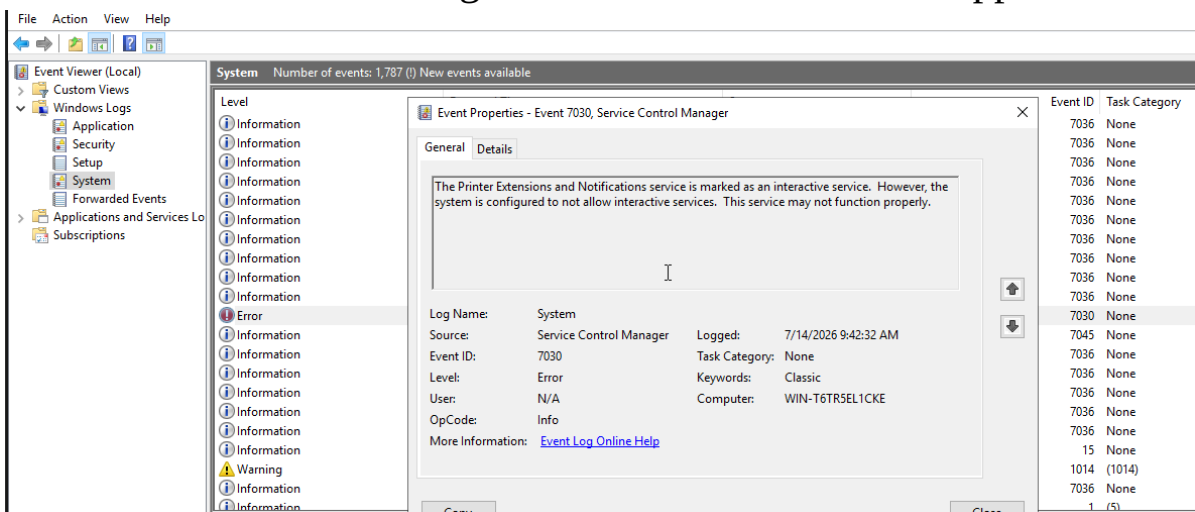
4. BitLocker Drive Encryption: helps to protect data on lost, stolen or inappropriately decommissioned computers by encrypting the entire volume and checking the integrity of early boot components. Data is only decrypted if those components are successfully verified and the encrypted drive is located in the original computer. Integrity checking requires a compatible Trusted Platform Module (TPM).



5. Event viewer



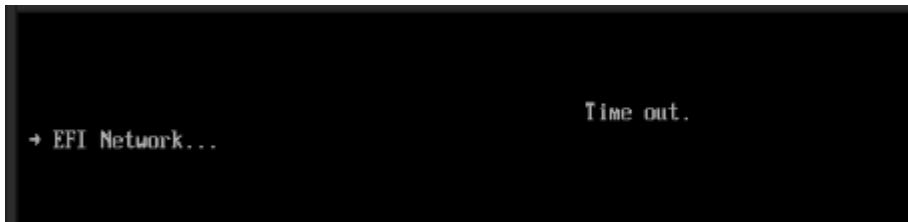
* mismatch was detected between the computer current name and the name recorded in its log files associated with CRM application.



* Printer Extensions and Notifications service is trying to open a user-interface window, but the Windows Server operating system is configured to block services from interacting with the user interface.

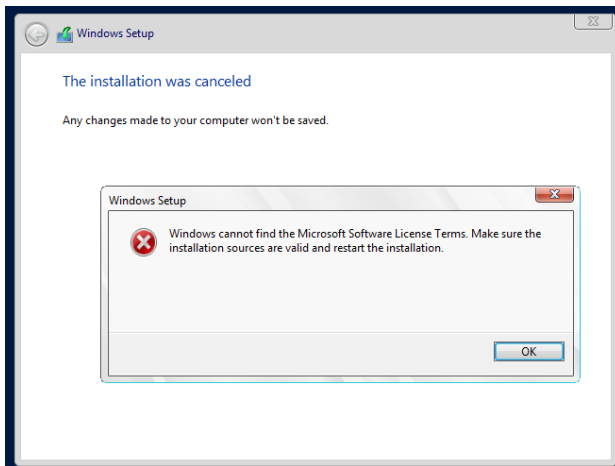
Challenges faced

1. VM failing to boot from the installation media (the ISO file) and is instead trying to boot from the network (PXE boot), which is timing out.



Solution: enter BIOS/EFI menu and force to use CDROM/ISO file

2. ISO file not working



Solution attempts:

- Re-mount the ISO file from settings (didn't solve the problem)
- Verify if ISO file is corrupted comparing has values -> Linux command 'sha256sum <file.iso'

https://github.com/ruzickap/packer-templates/blob/main/win_iso.sha256

<https://gitlab.com/badsectorlabs/ludus/-/blob/1.11.6/templates/win2019-server-x64-no-security-updates/win2019-server-x64-no-security-updates.pkr.hcl>

(hash is correct, problem still stands)

- Remove floppy disk: automated floppy disk causes this problem because it introduces a configuration file (autoinst.flp) that attempts to "pre-seed" the Windows installation.

(problem solved)

3. Server VM can't connect to target VM via ip address (firewall settings problem)

```
C:\Users\Administrator>ping 192.168.203.129

Pinging 192.168.203.129 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.203.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

fix: Windows Defender Firewall with Advanced Security -> inbound rules -> enable: File and Printer Sharing (Echo Request - ICMPv4-In) private rule

File and Printer Sharing (Echo Request - ICMPv4-In)	File and Printer Sharing	Domain	No	Allow
✓ File and Printer Sharing (Echo Request - ICMPv4-In)	File and Printer Sharing	Private...	Yes	Allow
File and Printer Sharing (Echo Request - ICMPv6-In)	File and Printer Sharing	Domain	No	Allow

Learning outcomes

- Installing and configuring Windows Server 2019 VM

- Initial server administration using Server Manager.
- Installing Windows features such as BitLocker Drive Encryption.
- Use Device Manager, Services, and Firewall settings to troubleshoot common server issues.
- Verifying network connectivity using command-line utilities including ipconfig, ping, tracert, nslookup, and hostname.
- Analyze Application, System, and Security logs using Event Viewer to identify warnings and errors.

Conclusion

The basis for effectively and securely managing business IT infrastructure is provided by Windows Server administration. System administrators can maintain dependable, safe, and well-managed server environments by having a solid understanding of server deployment, initial configuration, system customization, security features, remote administration, troubleshooting, and event monitoring.

References

- <https://www.geeksforgeeks.org/blogs/how-to-create-a-new->
- <https://learn.microsoft.com/en-us/windows-server/get-started/overview>
- <https://learn.microsoft.com/en-us/windows-server/administration/server-manager/server-manager>
- <https://www.ninjaone.com/blog/what-is-windows-device-manager/>